Notes: Grossman and Hart (Econometrica, 1983)

An Analysis of the Principal-Agent Problem

Contents

1 Big picture

- **Question.** When a principal cannot observe the agent's action, what is the optimal incentive scheme, and how can we characterize the welfare loss from this moral-hazard problem?
- **Setting.** A principal chooses a contract that maps observable outcomes into payments. The agent privately chooses an action that affects the probability distribution over outcomes.
- **Core challenge.** Much of the earlier literature used a **local first-order approach**: make the agent indifferent at the margin and optimize subject to the agent's first-order condition. Mirrlees showed this can fail when the agent has multiple global optima.
- **Main methodological result.** Under the assumption that the agent's preferences over income lotteries are independent of action, the problem of implementing any target action can be rewritten as a **convex programming problem**.
- **Main characterization results.** The paper derives existence of an optimal second-best contract, shows that first-best and second-best generally differ, characterizes when optimal contracts are monotone/progressive/regressive, proves that better information in the Blackwell sense reduces implementation cost, and obtains sharp results in the two-outcome case.
- **Key economic message.** The principal-agent problem is not just about risk sharing. Observable outcomes serve both as consumption and as **signals** of hidden action. These two roles can conflict, which is why optimal incentive schemes can be subtle and need not be globally monotone without strong assumptions.
- **Why this paper matters.** This is a foundational paper because it replaces an invalid local method with a rigorous global method and shows exactly what can and cannot be inferred about optimal incentive contracts.

2 Data and measurement (key objects)

This is a **theory paper**; there is no empirical dataset. The "objects" are the primitives of the contract problem.

2.1 Observed economic environment

- **Outcomes.** There are finitely many observable gross-profit outcomes

$$q_1 < q_2 < \cdots < q_n.$$

- **Actions.** The agent chooses an unobservable action

$$a \in A,$$

  where A is a nonempty compact subset of a finite-dimensional Euclidean space.

- **Technology / signal structure.** Each action induces a probability vector over outcomes,

$$\pi(a) = \big(\pi_1(a), \ldots, \pi_n(a)\big), \qquad \pi_i(a) > 0, \quad \sum_{i=1}^{n} \pi_i(a) = 1.$$

- **Contract.** An incentive scheme is a vector of payments

$$I = (I_1, \dots, I_n),$$

where I_i is paid if outcome q_i is observed.

2.2 Preferences and first-best cost objects

- **Principal.** The principal is risk neutral and cares about expected net profit.
- **Agent.** The agent has von Neumann-Morgenstern utility

$$U(a, I) = G(a) + K(a)V(I),$$

with V strictly increasing and concave, and $K(a) > 0$.

- **Reservation utility.** To avoid the paper's notational clash between the utility function $U(a, I)$ and the reservation utility U, write the latter as $\bar{U}$.
- **Inverse utility transform.** Let

$$h \equiv V^{-1}.$$

Then first-best cost is

$$C^{FB}(a) = h\left(\frac{\bar{U} - G(a)}{K(a)}\right).$$

- **Cost ranking.** The function $C^{FB}(a)$ induces an ordering of actions by how costly they are to induce when action is observable.

2.3 Second-best objects

- **Implementation cost.** For each target action a, the second-best cost $C(a)$ is the minimum expected payment needed to make the agent choose a when actions are hidden.
- **Benefit.** Expected output benefit is

$$B(a) = \sum_{i=1}^{n} \pi_i(a) q_i.$$

- **Loss from hidden action.** The welfare loss to the principal is

$$L = \max_{a \in A}\big[B(a) - C^{FB}(a)\big] - \sup_{a \in A}\big[B(a) - C(a)\big].$$

- **Two-outcome notation.** When $n = 2$, the contract can be summarized by a fixed payment w and a profit share s:

$$I_1 = w + sq_1, \qquad I_2 = w + sq_2.$$

- **Information structure.** A noisier signal can be represented as a Blackwell garbling of the original signal,

$$\pi'(a) = R\pi(a),$$

where R is a stochastic matrix.

2.4 Timing and observability

- The principal offers an incentive scheme I.
- The agent decides whether to participate.
- If he participates, he privately chooses $a \in A$.
- An outcome q_i is realized according to $\pi(a)$.
- The principal observes the outcome, not the action, and pays I_i.
- The paper assumes that if the agent is indifferent across actions, he breaks ties in the principal's favor. This matters for implementation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Principal's payoff and agent's hidden action

If the principal wants to induce action a, expected gross benefit is

$$B(a) = \sum_{i=1}^{n} \pi_i(a) q_i.$$

If the principal pays I_i in state i, expected net payoff is

$$\sum_{i=1}^{n} \pi_i(a) \big(q_i - I_i\big).$$

- **Interpretation.** Outcome q_i matters twice: it affects the principal's consumption directly, and it is also the signal used to create incentives.
- **Why this matters.** A state that is desirable for production may be undesirable as an incentive state, and vice versa.

3.2 Assumption A1 and the structure of preferences

The key assumption is

$$U(a, I) = G(a) + K(a)V(I).$$

- **Interpretation.** Preferences over income lotteries are independent of the chosen action.
- **Special cases.**
  - **Additive separability:** $K(a)$ constant.
  - **Multiplicative separability:** $G(a) = 0$.
- **Example.** If $V(I) = -e^{-kI}$ and $K(a) = e^{ka}$, then

  $$U(a, I) = -e^{-k(I-a)},$$

  so effort acts like negative income.

3.3 First-best benchmark

If action is observable, the principal can make payment contingent directly on a. The minimum payment that delivers reservation utility $\bar{U}$ is

$$C^{FB}(a) = h\left(\frac{\bar{U} - G(a)}{K(a)}\right).$$

The first-best action solves

$$\max_{a \in A} \big[B(a) - C^{FB}(a)\big].$$

- **Interpretation.** $C^{FB}(a)$ is the agent's reservation price for taking action a when action can be enforced directly.
- **Cost ordering.** Higher $C^{FB}(a)$ means a costlier action for the agent.

3.4 Second-best implementation problem in levels

Fix a target action a^*. The principal solves

$$\min_{I_1,\dots,I_n} \sum_{i=1}^{n} \pi_i(a^*) I_i \tag{2.1}$$

subject to

$$\sum_{i=1}^{n} \pi_i(a^*) U(a^*, I_i) \geq \bar{U},$$

and

$$\sum_{i=1}^{n} \pi_i(a^*) U(a^*, I_i) \geq \sum_{i=1}^{n} \pi_i(a) U(a, I_i) \qquad \text{for all } a \in A.$$

- **Participation.** The agent must weakly prefer working for the principal.
- **Global incentive compatibility.** The target action must dominate *every* other action, not just nearby deviations.
- **Why the paper is needed.** This global condition is exactly what the local first-order method does not handle correctly.

3.5 Convex programming reformulation

Let

$$v_i = V(I_i), \qquad h = V^{-1}.$$

Then the implementation problem becomes

$$\min_{v_1,\dots,v_n} \sum_{i=1}^{n} \pi_i(a^*) h(v_i) \tag{2.2}$$

subject to

$$G(a^*) + K(a^*) \sum_{i=1}^{n} \pi_i(a^*) v_i \geq G(a) + K(a) \sum_{i=1}^{n} \pi_i(a) v_i \qquad \text{for all } a \in A,$$

and

$$G(a^*) + K(a^*) \sum_{i=1}^{n} \pi_i(a^*) v_i \geq \bar{U}.$$

- **Key result.** The constraints are linear in the transformed variables v_i, and the objective is convex because h is convex.
- **Why this is the main technical contribution.** Under A1 the problem is globally tractable. Without A1, this convexification generally fails.
- **Definition of implementation.** If $(v_1, \ldots, v_n)$ satisfies these constraints, it implements a^*.

3.6 Cost-benefit decomposition and existence

Define the second-best cost function as

$$C(a) = \inf \left\{ \sum_{i=1}^{n} \pi_i(a) h(v_i) : (v_1, \ldots, v_n) \text{ implements } a \right\},$$

with $C(a) = \infty$ if a cannot be implemented.

The principal then solves the second stage

$$\max_{a \in A} \big[B(a) - C(a) \big].$$

Under assumptions A1–A3, Proposition 1 establishes existence of a second-best optimal action and an optimal incentive scheme.

Additional benchmark results are:

- If utility is additively or multiplicatively separable, the participation constraint binds at the optimum (Proposition 2).
- In general,
$$C(a) \geq C^{FB}(a),$$
so the second-best is weakly costlier than the first-best (Proposition 3).
- Equality holds only in special cases: for example when V is linear, or when the first-best action is already the least costly action, or when some outcomes perfectly identify the intended action.
- **Interpretation.** Moral hazard creates a wedge between technological efficiency and contractual feasibility.
- **Economic meaning of A3.** Positive probabilities for all outcomes rule out cases where the principal can approach implementation through arbitrarily severe punishments on events of arbitrarily small probability.

3.7 Why global incentive compatibility changes the shape results

A central lemma of the paper compares two optimal contracts that leave the agent with the same expected utility but induce different actions. This leads to Proposition 4:

> It is not optimal for the principal's and agent's payoffs to be negatively related over the *entire* outcome range.

Formally, the paper shows it cannot be true that

$$I_i > I_j \iff q_i - I_i < q_j - I_j$$

for all states, with strict reversal somewhere.

Proposition 5 then implies:

- there exists some adjacent pair with
$$I_i < I_{i+1},$$
unless the contract is completely flat;
- and there exists some adjacent pair with
$$q_j - I_j < q_{j+1} - I_{j+1}.$$
- **Interpretation.** The contract cannot be globally "backward sloping" for the agent, and the principal's own payoff cannot be globally backward sloping either.
- **But the paper is careful.** This is weaker than saying the whole contract must be monotone. The informational role of outcomes can still justify local nonmonotonicity.

3.8 Kuhn-Tucker characterization and binding deviations

When A is finite and h is differentiable, the Kuhn-Tucker conditions for implementing a^* imply

$$h'(v_i) = \left[\lambda + \sum_{a_j \neq a^*} \mu_j\right] K(a^*) - \sum_{a_j \neq a^*} \mu_j K(a_j) \frac{\pi_i(a_j)}{\pi_i(a^*)}, \tag{3.9}$$

where $\lambda \geq 0$ is the multiplier on participation and $\mu_j \geq 0$ is the multiplier on the incentive constraint for action a_j.

Proposition 6 shows that if the target action is not the least costly one, then at the optimum the agent must be indifferent between a^* and at least one **less costly** action. In other words, some $\mu_j > 0$ for an action a_j with

$$C^{FB}(a_j) < C^{FB}(a^*).$$

- **Interpretation.** The relevant deviations are not infinitesimal. Optimal contracts are often pinned down by discrete alternative actions that are just attractive enough to bind.
- **Why this matters for the literature.** This is the mechanism behind the failure of the local first-order approach.

3.9 Monotonicity: MLRC is not enough

The monotone likelihood ratio condition (MLRC) is

$$C^{FB}(a') < C^{FB}(a) \quad \Longrightarrow \quad \frac{\pi_i(a')}{\pi_i(a)} \text{ is nonincreasing in } i.$$

The local first-order literature suggests MLRC should be enough for a monotone optimal contract. Grossman and Hart show it is **not** enough once multiple actions can bind globally. Their Example 1 satisfies MLRC yet yields an optimal contract that is not nondecreasing.

- **Why MLRC fails by itself.** If the target action is simultaneously tied with a cheaper action and a more expensive action, the weighted combination of their likelihood ratios need not be monotone.
- **Takeaway.** Global incentive compatibility requires stronger conditions than the local method suggests.

3.10 Stronger conditions for monotonicity, progressivity, and regressivity

The paper offers two ways to recover monotonicity.

First, a strong **spanning condition** (SC): there exist two probability vectors π^L and π^H such that every action's outcome distribution can be written as

$$\pi(a) = \lambda(a)\pi^L + [1 - \lambda(a)]\pi^H, \qquad 0 \le \lambda(a) \le 1,$$

and the likelihood ratio π_i^L/π_i^H is nonincreasing in i.

Second, under additive or multiplicative separability, MLRC plus a **concavity of distribution function condition** (CDFC) are sufficient. Let

$$F(a) = \big(\pi_1(a),\ \pi_1(a) + \pi_2(a),\ \ldots,\ \pi_1(a) + \cdots + \pi_{n-1}(a)\big).$$

CDFC requires that if action a is a convex combination of two other actions in terms of first-best cost, then its cumulative distribution vector $F(a)$ dominates the corresponding convex combination.

Under SC (Proposition 7) or MLRC + CDFC under separability (Proposition 8), the optimal contract is monotone:

$$I_1 \le I_2 \le \cdots \le I_n.$$

For curvature of the contract, Proposition 9 shows:

- the contract is **regressive** if $1/V'(I)$ is concave and the likelihood-ratio increments satisfy a monotone-difference condition;
- the contract is **progressive** if $1/V'(I)$ is convex and the same likelihood-ratio increments move the other way.

- **Interpretation.** Monotonicity itself is hard; curvature is even harder. The paper gives sufficient conditions, not generic conclusions.

3.11 The two-outcome case: linear sharing and efficient actions

When $n = 2$, any contract can be written as

$$I_1 = w + sq_1, \qquad I_2 = w + sq_2,$$

so s is the agent's share of profit and w is a fixed payment.

The paper derives several sharp results.

- Proposition 5 implies
$$0 < s < 1.$$
So the optimal contract is neither fully flat ($s = 0$) nor full residual claimancy ($s = 1$).
- Proposition 10: every second-best optimal action is **efficient**. In the two-outcome case, the principal never wants to induce an action that is dominated by a cheaper action with at least as high a probability of the good outcome.
- Proposition 11: participation binds even beyond the additive/multiplicative separable cases. Once s is chosen, the fixed payment w is pinned down by
$$\max_{a \in A} \Big\{ G(a) + K(a) \big[\pi_1(a) V(w + sq_1) + \pi_2(a) V(w + sq_2) \big] \Big\} = \bar{U}.$$
- Proposition 12: if the action set expands only by adding actions that are *more costly* than the original actions, the principal cannot be made worse off.
- **Interpretation.** With only two outcomes the geometry becomes much cleaner, and the paper obtains much stronger qualitative results.

3.12 Computational algorithm in the two-outcome case

Suppose A is finite and efficient actions are ordered so that

$$C^{FB}(a_1) < C^{FB}(a_2) < \cdots < C^{FB}(a_m), \qquad \pi_2(a_1) < \pi_2(a_2) < \cdots < \pi_2(a_m).$$

Then:

- for the cheapest action,
$$C(a_1) = C^{FB}(a_1);$$
- for $k > 1$, for each $j < k$, solve
$$G(a_k) + K(a_k) \big[\pi_1(a_k) v_1 + \pi_2(a_k) v_2 \big] = \bar{U}, \tag{4.4a}$$
$$G(a_j) + K(a_j) \big[\pi_1(a_j) v_1 + \pi_2(a_j) v_2 \big] = \bar{U}. \tag{4.4b}$$

This gives

$$v_1 = \frac{\pi_2(a_j) \frac{\bar{U} - G(a_k)}{K(a_k)} - \pi_2(a_k) \frac{\bar{U} - G(a_j)}{K(a_j)}}{\pi_1(a_k) - \pi_1(a_j)}, \tag{4.5a}$$

$$v_2 = \frac{\pi_1(a_j)\frac{\bar{U}-G(a_k)}{K(a_k)} - \pi_1(a_k)\frac{\bar{U}-G(a_j)}{K(a_j)}}{\pi_2(a_k) - \pi_2(a_j)}. \tag{4.5b}$$

Set

$$I_1 = h(v_1), \qquad I_2 = h(v_2).$$

Among all such candidate pairs, choose the one with the **smallest** v_1 (equivalently the largest v_2). Then

$$C(a_k) = \pi_1(a_k)h(v_1) + \pi_2(a_k)h(v_2).$$

Finally compare

$$B(a_k) - C(a_k)$$

across $k = 1, \dots, m$.

- **Interpretation.** Figure 2 shows the geometry. For each target action a_k, candidate contracts are generated by making the agent indifferent between a_k and one cheaper action. The true solution is the candidate with the smallest bad-state utility.

3.13 What determines how serious the incentive problem is?

Information quality. If the new information structure is a garbling of the old one,

$$\pi'(a) = R\pi(a),$$

with R stochastic, then Proposition 13 shows

$$C'(a) \geq C(a) \qquad \text{for all } a \in A.$$

If gross output is preserved in expectation, so that

$$q'R = q,$$

then the loss from hidden action rises:

$$L' \geq L.$$

Repeated garbling drives the loss to its maximal value L^* (Proposition 14).

- **Interpretation.** Worse signals about hidden action make incentives more expensive. In the limit, the principal loses effective control and the agent drifts to the least-cost action.

Risk aversion. For the special CARA family

$$U(a, I) = -e^{-k(I-a)}, \qquad \bar{U} = -e^{-k\alpha},$$

first-best cost is

$$C^{FB}(a) = \alpha + a,$$

which does not depend on k. Proposition 15 shows

$$\lim_{k \to 0} L(k) = 0, \qquad \lim_{k \to \infty} L(k) = L^*.$$

With $n = 2$ and finite A, Proposition 16 shows that $L(k)$ is increasing in k.

- **Interpretation.** The more risk averse the agent, the more costly it becomes to use outcome-contingent pay to provide incentives.

Incremental action costs. Proposition 17 studies a family of additive-separable problems in which the marginal disutility of inducing action is scaled by a parameter λ. Under regularity conditions,

$$\lim_{\lambda \to 0} \frac{L(\lambda)}{\lambda} = 0.$$

- **Interpretation.** When the incentive problem is small, the second-best welfare loss is of smaller than first-order magnitude.

4 Identification and instruments (what makes the estimates credible)

This is a **theory paper**, so there are no empirical instruments. The identification issue is instead: what assumptions are doing the work in turning the principal's global problem into something tractable and in making the comparative statics valid?

4.1 What fails in the standard approach

- The standard local method imposes the agent's first-order condition and ignores the possibility of multiple global optima.
- Mirrlees showed this is generally invalid.
- Figure 1 in the paper is the cleanest intuition: the true feasible set is the set of global optima, not the entire locus of stationary points.

4.2 What identifies the new convex-programming approach

- **A1** gives the separable utility structure needed to transform the contract problem into the $v_i = V(I_i)$ space.
- Concavity of V implies convexity of $h = V^{-1}$, which makes the objective convex.
- The transformed incentive and participation constraints become linear, so the problem is globally well behaved.
- The tie-breaking assumption means implementation is defined even when several actions give the agent the same utility.

4.3 Why existence and global characterization need A3

- A3 requires every outcome to occur with strictly positive probability under every action.
- This rules out sequences of contracts that punish vanishing-probability events more and more harshly, which could otherwise make optima fail to exist.
- A3 is therefore the key regularity condition behind Proposition 1.

4.4 What identifies monotonicity and curvature results

- MLRC alone is not enough because multiple global incentive constraints can bind.
- SC or MLRC + CDFC restore enough structure on the family of outcome distributions to recover monotonicity.
- Additional curvature restrictions on $1/V'(I)$ and on likelihood-ratio differences identify when the optimal schedule is progressive or regressive.

4.5 What identifies comparative statics on the seriousness of the problem

- Blackwell garblings identify "worse information" in a formally comparable way.
- The CARA family parameter k gives a disciplined way to vary risk aversion while preserving the utility form needed for analysis.
- The small-λ family in Proposition 17 gives a disciplined way to scale the size of the action-cost problem.

4.6 Relation to the literature

- Relative to Mirrlees, Holmstrom, and others, the paper's main difference is methodological: it solves the **global** implementation problem instead of optimizing around the agent's local first-order condition.
- Relative to the general incentive-compatibility literature, this paper studies a pure monitoring problem rather than a private-information message problem.
- The paper also foreshadows later work on informativeness and Blackwell comparisons by showing directly how noisier signals raise implementation costs.

5 Tables and figures

The paper is a **theory paper**: it has **two figures and no empirical tables**. The real content is carried by the figures and the sequence of propositions.

5.1 Figure 1: Why the standard first-order method fails

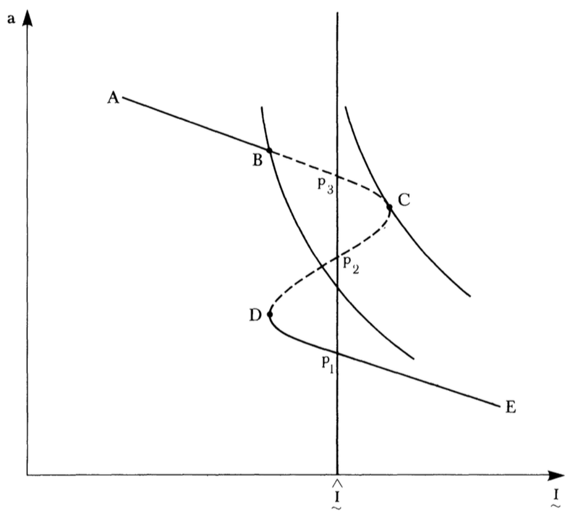
Read:

- The curve of stationary points for the agent is larger than the set of **global** maxima.
- The principal's true feasible set consists only of contracts/actions at which the agent globally optimizes.
- A point that solves the relaxed "agent FOC" problem can be infeasible for the true problem.
- This figure is the paper's opening conceptual contribution: local stationarity is not enough.

5.2 Propositions 1–3: existence and the wedge between first-best and second-best

Read:

- Proposition 1 establishes existence of a second-best optimal action and contract under A1–A3.
- Proposition 2 shows participation binds in the additive and multiplicative separable cases.
- Proposition 3 shows second-best implementation is generally more costly than first-best, with equality only in special benchmark cases.
- This block tells you the basic welfare geometry of the model: hidden action typically creates a real wedge.



For a given $\underset{\sim}{I}$ the agent strictly prefers lower actions

FIGURE 1.

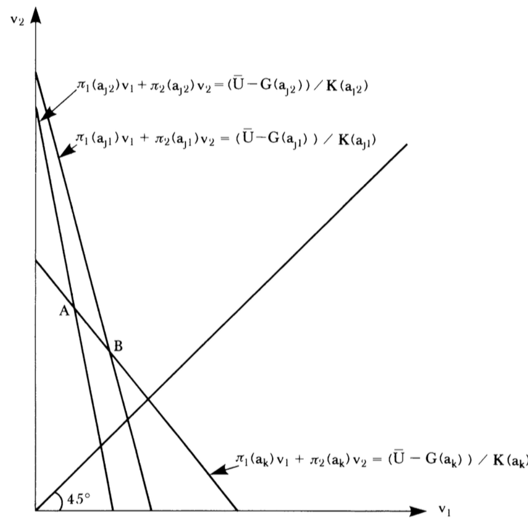


FIGURE 2.

5.3 Propositions 4–9 and Example 1: the shape of optimal incentive schemes

Read:

- Proposition 4 says the principal and agent cannot have payoffs moving in opposite directions over the whole outcome range.
- Proposition 5 says some local positive relation must exist in both the agent's compensation and the principal's net payoff.
- Proposition 6 shows binding deviations come from **cheaper actions**, which explains why local FOCs are incomplete.
- Example 1 is crucial: MLRC alone does *not* guarantee monotonicity once several actions can bind.
- Propositions 7–9 then show what extra structure is strong enough to recover monotonicity and curvature.

5.4 Figure 2 and Propositions 10–12: the two-outcome case

Read:

- Figure 2 gives the geometric algorithm for computing the optimal two-state contract.
- Proposition 10 says second-best optimal actions are efficient when there are only two outcomes.
- Proposition 11 says the agent's participation constraint binds, so the fixed payment is pinned down once the share is known.
- Proposition 12 says adding only higher-cost actions cannot hurt the principal.
- This is the cleanest part of the paper: two outcomes turn the general contract problem into a tractable linear-sharing problem.

5.5 Propositions 13–17: what makes the incentive problem more serious

Read:

- Proposition 13 says Blackwell-garbled information raises implementation cost action by action.
- Proposition 14 says repeated worsening of information drives the welfare loss up to its maximal value.
- Propositions 15 and 16 show that greater agent risk aversion makes the incentive problem more severe in the CARA case, especially sharply in the two-outcome finite-action case.
- Proposition 17 shows that when the incentive problem itself becomes small, welfare losses are smaller than first-order.
- This block is the paper's comparative-statics payoff: it explains which features of the environment make moral hazard quantitatively important.

6 Short summary

- **Conclusion.** Grossman and Hart replace an invalid local method for the principal-agent problem with a global convex-programming approach.
- **Core insight.** The key to tractability is the assumption that the agent's preferences over income lotteries are independent of action, which lets the implementation problem be solved in transformed utility space.
- **Main lesson.** Optimal contracts are shaped by global incentive compatibility, not by local first-order conditions; better information and lower risk aversion make the incentive problem less severe.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides a rigorous way to solve the principal-agent problem when actions are hidden and the agent's lottery preferences are action independent.
- It shows that the least-cost way to implement any given action is a convex program, so the principal's problem can be decomposed into a cost side and a benefit side.
- It then uses that decomposition to characterize the structure of optimal incentive schemes and the determinants of second-best welfare loss.

7.2 Economic intuition

- Incentives require the principal to make payments vary with outcomes, but outcome variation also exposes the agent to risk.
- Because outcomes are both payoffs and signals, there is no reason for the optimal contract to be globally simple unless strong assumptions are imposed on the distributional structure of outcomes.
- The right comparison is therefore not "risk sharing versus incentives" in the abstract, but **which deviations bind globally** and how informative the observable signal is about hidden action.

7.3 Takeaway

This paper is best read as a methodological reset for the early principal-agent literature. Its deepest point is that the moral-hazard problem is a *global* optimization problem. Once one respects that fact, the geometry of the contract changes: local first-order conditions are not enough, monotonicity is not automatic, and information quality becomes central. The payoff is a theory that is both more rigorous and more economically revealing.